



SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS 28 / 31 completed

Arithmetic

ARITHMETIC
Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR Java Run Save

```
1 import java.util.*;
2 public class first
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int a=sc.nextInt();
8         int b=sc.nextInt();
9         System.out.println("sum:"+a+b);
10        System.out.println("difference:"+a-b);
11        System.out.println("product:"+a*b);
12        if(b!=0){
13            System.out.println("quotient:"+a/b);
14        }
15        else {
16            System.out.println("given number not is quotient");
17        }
18    }
19 }
20
```

INPUT

5
5

OUTPUT
sum:10
difference:0
product:25
quotient:1

QUESTIONS 28 / 31 completed

MAX

MAX
Write a Java program to find the maximum of three numbers using conditional statements.

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```
1 class main{
2     public static void main(String[] args)
3     {
4         int a=10;
5         int b=25;
6         int c=9;
7         if(a>=b && a>=c){
8             System.out.println("maximum:"+a);
9         }
10        else if(b>=a && b>=c){
11            System.out.println("maximum:"+b);
12        }
13        else{
14            System.out.println("maximum:"+c);
15        }
16    }
17 }
```

OUTPUT

```
maximum:25
```

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

```
1 class main{
2   public static void main(String[] args)
3   {
4     int year=2024;
5     if((year%400==0)|| (year%4==0 && year%100!=0)){
6       System.out.println("leap year");
7     }
8     else{
9       System.out.println("not leap year");
10    }
11  }
12 }
```

OUTPUT

```
leap year
```

QUESTIONS 28 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 public class main
2 {
3   public static void main(String[] args)
4   {
5     int n=10;
6     int first=0,second=1;
7     System.out.print("fibinacci series:");
8     for(int i=1;i<=n;i++){
9       int next=first+second;
10      first=second;
11      second=next;
12      System.out.println(" "+next);
13    }
14  }
15 }
16 }
```

OUTPUT

```
fibonacci series: 1
2
3
5
8
13
21
34
55
89
```

QUESTIONS 28 / 31 completed

RT-Pattern ▾

RT-PATTERN

Write a Java program to print the following pattern using loops:

```
**
***
****
*****
```

PROGRAM EDITOR

Java ▾

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         for(int i=1;i<=5;i++){
5             for(int j=1;j<=i;j++){
6                 System.out.print("*");
7             }
8             System.out.println();
9         }
10    }
11 }
```

OUTPUT

```
*
**
***
****
*****
```

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

```
1 public class gcd
2 {
3     public static void main(String[] args)
4     {
5         int a=56;
6         int b=98;
7         while(b!=0){
8             int temp=b;
9             b=a%b;
10            a=temp;
11        }
12        System.out.println("gcd:"+a);
13    }
14 }
```

OUTPUT

OUTPUT

gcd:14

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

```
1 public class first
2 {
3     public static void main(String[] args)
4     {
5         int number=27;
6         boolean isPrime = true;
7         if(number<=1){
8             isPrime = false;
9         }
10        else{
11            for(int i=2;i<=number;i++){
12                if(number%i==0){
13                    isPrime = false;
14                    break;
15                }
16            }
17        }
18        if(isPrime){
19            System.out.println(number+"is a prime number");
20        }
21        else{
```

```
19     System.out.println(number+"is a prime number");
20 }
21 else{
22     System.out.println(number+"is not a prime number");
23 }
24 }
25 }
```

OUTPUT

```
27is not a prime number
```

QUESTIONS 28 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run

Save

```
1 public class factorial
2 {
3     static int fact(int x)
4     {
5         if(x==0 || x==1){
6             return 1;
7         }
8         else{
9             return x*fact(x-1);
10        }
11    }
12 }
13 public static void main(String[] args)
14 {
15     int n=5;
16     System.out.println(fact(n));
17 }
18 }
```

OUTPUT

OUTPUT

120

QUESTIONS 28 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class lcm
2 {
3     public static void main(String[] args){
4         int num1=15;
5         int num2=20;
6         int a=num1;
7         int b=num2;
8         while(b!=0){
9             int temp=b;
10            b=a%b;
11            a=temp;
12        }
13        int gcd=a;
14        int lcm=(num1*num2)/gcd;
15        System.out.println("lcm:"+lcm);
16    }
17 }
```

OUTPUT

lcm: 60

QUESTIONS 28 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 class main{
3     public static void main(String[] args)
4     {
5         Scanner sc=new Scanner(System.in);
6         System.out.print("enter a number:");
7         int num=sc.nextInt();
8         int dup=num;
9         int rev=0;
10        while(num>0){
11
12
13            rev=rev*10+num%10;
14
15            num=num/10;
16        }
17        if(dup==rev){
18            System.out.println("palindrome");
19        }
20        else{
21            System.out.println("not palindrome");
```

INPUT

121

OUTPUT

```
21         System.out.println("not palindrome");
22     }
23     System.out.println("reversed number = "+rev);
24 }
25 }
```

```
enter a number:palindrome
reversed number = 121
```

QUESTIONS 28 / 31 completed

RT-num.Pattern

RT- NUM.PATTERN

Write a Java program to print the following pattern using loops:

```
1
22
333
4444
55555
```

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         for(int i=1;i<=5;i++){
5             for(int j=1;j<=i;j++){
6                 System.out.print(i);
7             }
8             System.out.println();
9         }
10    }
11 }
```

OUTPUT

```
1
22
333
4444
55555
```

QUESTIONS 28 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class bubble
3 {
4     public static void main(String[] args)
5     {
6
7         int[] arr={5,3,8,1,2};
8         int n=arr.length;
9
10        for(int i=0;i<n-1;i++){
11            for(int j=0;j<n-1-i;j++){
12                if(arr[j]>arr[j+1]){
13                    int temp=arr[j];
14                    arr[j]=arr[j+1];
15                    arr[j+1]=temp;
16                }
17            }
18        }
19        for(int i=0;i<n;i++){
20            System.out.println(arr[i]+" ");
21        }
```

OUTPUT

```
1
2
3
5
8
```